Where the Admissible Radius Binds: a Correction and a Measured Boundary in Online Conformal Prediction

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Abstract

A printed claim that the scorecaster breaks online conformal coverage meets its refutation by overlapping authors; this paper joins them and locates the edge of the set that refutation relies on. Conformal PID states long-run coverage for any bounded scorecaster; a corollary keeps it for predictable perturbations of the deployed threshold inside an admissible radius. The claim was acted on; placement and derivation are conceded by name. At the null scorecaster that condition is *tight*. A legal L_1 dead band on the *completed* threshold leaves that set at its radius, and past it miscoverage is 1.000000 against $\alpha = 0.1$. Coverage goes, not only the rate. The edge belongs to the saturator–scorecaster pair, not the readout; elsewhere the condition is sufficient only. Under the equally legal $\hat{q} \equiv +b/2$ the failing band covers; at $\hat{q} \equiv -b/2$ both it and partial adjustment at $w = 0.999$ miscover.

1 Introduction

An online conformal method emits a threshold before each observation and moves it. Forecast stability names, scores and prices it [Tunc et al., 2013, Godahewa et al., 2025, Van Belle et al., 2023, Pritularga and Kourentzes, 2024, Genov et al., 2026, Van Belle et al., 2026]. Varying temporal structure at fixed level came earlier, in forecast verification, matching predictive marginals on real producers [Gneiting et al., 2007, Pinson and Girard, 2012, Worsnop et al., 2018]. Coverage and mean length under-describe a deployed interval [Min et al., 2026, Vaze, 2026]. None of it is claimed here.

First: the record carries a claim and its refutation, and this paper joins them. Conformal PID [Angelopoulos et al., 2023] adds a *scorecaster* $\hat{q}_{t+1}$ to a saturating error integrator, asking only that it be predictable and lie in $[-b/2, b/2]$: "any function of the past" (Theorem 1). A readout there is already quantified over. Also in print, of that method: "[s]ince using a scorecaster (D-part) in C-PID breaks the theoretical coverage guarantee" [Li and Rodríguez, 2025], stated and acted on in its baselines paragraph, verbatim on refereed printed page 18443. It is refuted twice, with proof. Li et al. [2025, Corollary 2] add an arbitrary predictable d_t with $|d_t| \leq \mu_t(b/2 - |\hat{q}_t|)$ to the deployed threshold, note that the result "has exactly the CPID error-integration form", and conclude $|E_T| = o(T)$. The same generic-slot argument carries AcMCP's long-run coverage [Wang and Hyndman, 2024]. The first and last authors of Li et al. [2025] wrote Li and Rodríguez [2025] and cite it: the halves already share a bibliography, this is ordinary self-correction, and the joining is what this paper adds. Four surfaces were queried on 20 August 2026 for both identifiers and "scorecaster". They are arXiv full text, arXiv abstracts, OpenReview, and the six works Semantic Scholar records citing Li and Rodríguez [2025]. None returns both the claim and the corollary. Hu et al. [2026] still call scorecaster choice "arbitrary" and lacking "principled guidance" while asserting valid coverage: model selection, not validity. **We claim neither the placement nor the derivation.**

Second: that admissible set has an edge, and the condition there is tight. Corollary 2 is a *retention* result: what it leaves open is outside. Section 3 exhibits a legal perturbation leaving it, the L_1 dead-band family characterised here on the *completed* threshold, whose edge is the corollary's

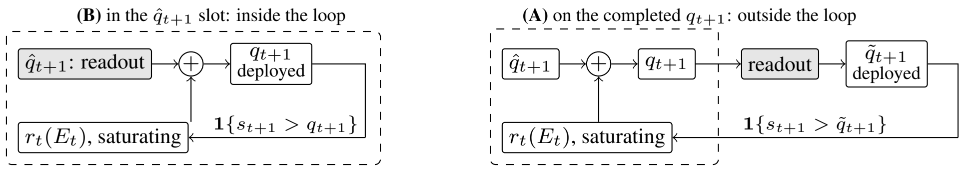

Figure 1: One scalar readout (grey) in its two places in (1); dashed is the loop r_t closes. In **(B)** the loop closes on the integrator's own output, and both readouts keep $\{\hat{q}_t\}$ in the convex hull of $\hat{q}_1$ and the raw scorecaster's range, inside Corollary 2's admissible set. In **(A)** it closes on $\tilde{q}_{t+1}$, which it never computed, and the deployed sequence leaves that set.

radius at the null scorecaster. Past it coverage goes, not only the rate: the published sufficient condition is necessary there. Section 4 sets both claims beside four vocabularies for the same movement.

2 Setup

The producer. A forecaster fit before t induces a score s_t, and a threshold q_t is emitted before y_t is revealed. The deployed set is $C_t = \{y : s_t(x_t, y) \le q_t\}$ and $\mathrm{err}_t = \mathbf{1}\{s_t(x_t, y_t) > q_t\}$. The target is long-run coverage, $T^{-1} \sum_{t \le T} \mathrm{err}_t = \alpha + o(1)$, with no model on the data. Conformal PID [Angelopoulos et al., 2023] manipulates the threshold on the score scale,

$$q_{t+1} \;=\; \hat{q}_{t+1} \;+\; r_t\Big(\textstyle\sum_{i \le t} (\mathrm{err}_i - \alpha)\Big), \tag{1}$$

their iteration (5). Its r_t saturates: condition (4) requires $|r_t(x)| \ge b$ with the sign of x once $|x| \ge c\,h(t)$, for h nonnegative, nondecreasing and sublinear. With $E_t = \sum_{i \le t}(\mathrm{err}_i - \alpha)$, Proposition 2 gives $|E_T| \le c\,h(T) + 1$, read on realised scores.

One scalar, two readouts, two places. Damping a sequentially updated scalar has two standard readouts, both prior work. A quadratic cost gives linear partial adjustment $\tilde{q}_{t+1} = (1-\lambda)\hat{q}^{\mathrm{raw}}_{t+1} + \lambda \tilde{q}_t$ [Gârleanu and Pedersen, 2013], published as post-processing [Godahewa et al., 2025] and applied to a deployed threshold [Binny and Dixit, 2025, Eq. 13]. A proportional cost gives a no-trade band, moving only when the target is more than τ away, then pulling back exactly τ [Constantinides, 1986, Davis and Norman, 1990]. Either readout has two positions in (1), drawn in Figure 1; **(A)** is measured here. *The dead band (L_1) is not a stylistic alternative to the partial-adjustment (L_2) form*; it is the family whose failure Section 3 characterises. Neither is safe by construction: L_2 forfeits Proposition 2's rate while covering, L_1 past $\tau^\star$ loses coverage, and at the legal $\hat{q} \equiv -b/2$ both lose it. What is tight is the radius, not a form.

The quantity, and the names it already has. We report the deployed threshold's *travel*, $\mathcal{T}_H = \sum_{t=2}^{H} |q_t - q_{t-1}|$, after actuator travel in process control. The arithmetic is not new. Over a predictive distribution $\sum |\Delta q|$ is the Multi-Quantile Change [Zanotti, 2026, v3, §3.4.1]; verification calls it *jumpiness*, a *convergence index*, *(in)consistency* [Zsótér et al., 2009, Ehret, 2010, Pappenberger et al., 2011]; applied control, IACER [Sarhadi, 2025]; dynamic-regret online learning, a path length. Table 2's caption names ten, so a fresh one is used here.

Harness. The harness fixes $\alpha = 0.1$ and $b = 2$. Its saturator $r_t(x) = b\,\mathrm{clip}(x/(c\,h(t)), -1, 1)$, with $c = 1$ and $h(t) = \log(t+2)$, meets (4) with equality. The scorecaster is null and legal, reducing (1) to the pure error integration Proposition 2 addresses. Neither choice is neutral: (4) is a lower bound met exactly here, and Theorem 1 permits scorecasters $b/2$ either side of null. The adversary is specified, not tie-broken. It plays $s_t = \mathrm{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$, $\varepsilon = 10^{-9}$, against the *deployed* $\tilde{q}_t$: the $\varepsilon \downarrow 0$ reading of "sets the score at the threshold", firing the strict indicator. Legality is Theorem 1's $[-b/2, b/2]$ on both sides, so $b/2 + b/2 = b$ is exactly tight.

3 The edge of the admissible set, measured

The retention result is published, and tested from outside. Li et al. [2025, Corollary 2] prove that a predictable d_t with $|d_t| \le \mu_t(b/2 - |\hat{q}_t|)$ on the deployed threshold keeps (1) in error-integration form and *retains* $|E_T| \le c\,h(T) + 1$: the radius is theirs. A downstream partial adjustment of weight

Table 1: Placement A: a one-scalar readout on the *completed* threshold; one deterministic adversary, one pass to $T = 10^6$. *All eleven arms run are listed*, so no covering arm is withheld; five exceed the bound at $T = 10^6$. Proposition 2's bound is 13.2061 at $T = 2{\times}10^5$ and 14.8155 at $T = 10^6$; "ratio" is $\max_t |E_t|$ over it, "Sat." the fraction of rounds on which (4) delivers its full $\pm b$. Every figure is the *primary* regime's, defined in Appendix A, which also gives the $T = 10^4$ excursions and the fitted excursion law.

Readout on the completed threshold	miscov. $2{\times}10^5$	miscov. 10^6	$\max_t \lvert E_t\rvert$ 10^6	ratio	Sat. 10^6	travel 10^6
none (control)	0.100035	0.100007	7.80	0.53	0.0000	28,311
partial adj. $w = 0.5$	0.100030	0.100007	7.70	0.52	0.0000	18,113
partial adj. $w = 0.9$	0.100030	0.100007	7.50	0.51	0.0000	4,081
partial adj. $w = 0.99$	0.100040	0.100006	63.00	4.25	0.0007	1,023
partial adj. $w = 0.999$	0.100035	**0.100004**	623.70	42.10	0.0097	202
growing time constant $p = 0.5$	0.100030	0.100006	12.60	0.85	0.0000	330
growing time constant $p = 0.9$	0.100030	0.100024	53.00	3.58	0.3181	16
running mean	0.099940	0.100010	49.50	3.34	0.3647	9
dead band $\tau = 0.5$	0.100035	0.100005	10.60	0.72	0.0000	2,045
dead band $\tau = 0.9$	0.100060	0.100012	13.20	0.89	0.0038	1,051
dead band $\tau = 1.5$	**1.000000**	**1.000000**	900,000	60,747	1.0000	0.5

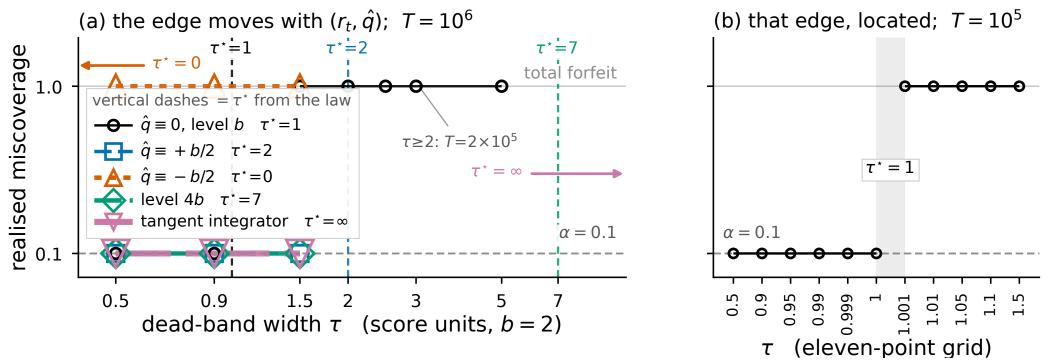

Figure 2: The dead-band coverage cliff, and the fact that its edge moves. **(a)** Realised miscoverage against dead-band width τ under five admissible $(r_t, \hat{q})$ pairs. Each pair's $\tau^\star = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$ is marked in its own colour, dashed where it falls on the axis and arrowed where it does not: 1 at the null scorecaster, 2 at $\hat{q} \equiv +b/2$ and 7 at a saturator of level $4b$ are dashed; 0 at $\hat{q} \equiv -b/2$ and unbounded under Angelopoulos et al.'s tangent integrator are arrows. Three widths run under each pair, plus four wider bands at the null scorecaster. All 19 dead-band runs land on the side of their own $\tau^\star$ that the law predicts, and the vertical dashes are the law's prediction, not four located edges: only the null scorecaster's edge has runs on both sides of $\tau^\star$. Appendix A gives the realised miscoverage at each covering setting. **(b)** An eleven-point grid at the null scorecaster *locates* that edge between $\tau = 1.000$ and 1.001, with no intermediate regime. Horizons: $T = 10^6$ in (a) except the four widest bands ($\tau \geq 2$), run only to $T = 2{\times}10^5$; $T = 10^5$ in (b).

w leaves it; Table 1 prices that departure. For $w = 0.99$ and 0.999 the excursion is *flat in* T while the bound grows like $\log T$, so the ratio falls with the horizon. At $w = 0.999$ it is 61.08 at $T = 10^4$ and 42.10 at $T = 10^6$. The forfeit is a property of the smoothing *strength*, not Placement A, and it buys something: travel falls from 28,311 to 202. The forfeit's sign is counter-intuitive. A readout on the completed q_{t+1} touches neither r_t nor the integrator's argument, so it cannot put (4) at risk. It delays a correction already computed, and the accumulator excurses *further*. Smoothing the output does not damp E_t; it feeds it. That 623.70 turns on the adversary's ε, not the counting convention.

The L_1 dead band leaves the radius, and the edge is measurable. A dead band pulls the deployed threshold back by exactly τ when it releases, so "inside Corollary 2" reads $\tau \leq \mu_t(b/2 - |\hat{q}_t|)$. Under saturation the deployed value is pinned τ below the integrator's ceiling, and its realised supremum is $b - \tau$ to the printed digit at every $\tau \geq 0.9$. The adversary's best legal play is $b/2$, so coverage is lost when that supremum drops below it, at $\tau^\star = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$. All 19 dead-band runs agree with that law (Figure 2); only the null scorecaster's edge is located, the others one-sided. **Here** $\sup_x r_t(x) = b$ **and** $\hat{q} \equiv 0$**, so** $\tau^\star = b/2$, one point of the admissible class, and both terms move it. The legal $\hat{q} \equiv +b/2$ carries the $\tau = 1.5$ band inside the bound. The equally legal $\hat{q} \equiv -b/2$ drops the edge to 0, and there $\tau = 0.5$ returns 1.000000, as does partial adjustment at $w = 0.999$. Condition (4) bounds $\sup_x r_t(x)$ only from below, so raising it costs the theorem nothing; it moves the edge too and shrinks the excursion (Figure 2). Under the unbounded tangent integrator Angelopoulos et al. [2023] report using in all their experiments, $\tau^\star$ is unbounded and all three widths there cover. The excursion is 20.10 rather than 623.70, both at $T = 10^6$. **The method's own integrator therefore sits inside the set at every width run here.** Outside the set the transition is a step. Past $\tau^\star$ the threshold sits permanently below the reachable score range and $|E_t| = (1-\alpha)t$ grows linearly, 60,747 times the bound at $T = 10^6$. The wider bands $\tau = 2.0$, 2.5, 3.0 and 5.0 fail too at $T = 2{\times}10^5$, so the failing set is $(\tau^\star, \infty)$, open at the left. **The endpoint is where the adversary's specification decides the answer.** At $\tau = \tau^\star$ the deployed supremum equals the adversary's own cap $b/2$, so the clip forces an exact play and the reading settles it: the primary regime covers, while scoring that play strictly returns 0.000000 and scoring it as a miscoverage returns 1.000000. Appendix A separates the two readings. **The failing arms have** $\mathrm{Sat.} = 1.0000$**.** Condition (4) delivers its full $\pm b$ on every round, and what fails is Proposition 2's other hypothesis: the threshold closing the loop *is* the integrator's output.

What that measures is that the published condition is tight. At $\mu_t = 1$, $\hat{q} \equiv 0$ the measured $\tau^\star$ is exactly Corollary 2's radius, and crossing it costs coverage rather than the rate. **The claim is that necessity: a tightness result about someone else's sufficient condition**, smaller and more precise than a forfeit of the rate by post-processing. Elsewhere it stays sufficient only: at $\hat{q} \equiv +b/2$ Corollary 2 permits radius 0 while every width run there covers. That asymmetry makes "tight" a located claim.

The constructive reading, and it is the corollary's rather than ours. Placement B keeps $\{\hat{q}_t\}$ inside the admissible radius, so Theorem 1 carries it already. What the measurement adds is that the band has an edge, computable before deployment from $\sup_x r_t(x)$ and $\sup_t \hat{q}_t$, and that crossing it is not a graceful degradation.

4 Where this sits

Four vocabularies name one quantity many ways, and the object that settles its validity (the admissible set around the deployed threshold) sits in the first of them. Table 2 in Appendix B names them and what each contributes. **What is conceded, by name.** The placement is prior work: Angelopoulos et al. [2023] state coverage for any admissible integrator and any scorecaster, and run a smoothing-family model in that slot themselves. Dupuy et al. [2026, Appendix A] give the generic argument, and Duerst et al. [2024] adopt that scorecaster. The derivation is prior work too, and this concession matters most. Li et al. [2025, Corollary 2] prove the retention statement in Conformal PID's own notation, with Wang and Hyndman [2024] an independent second instance. Section 2 concedes the smoother object, the metric arithmetic and both readout maps. Zheng et al. [2025], Wu et al. [2025], Chen et al. [2023], Lade et al. [2026] each place a smoother inside their own validity argument. One vocabulary out, this is integrator anti-windup with a feedforward path: free-until-the-bound-binds is its defining specification, not a finding [Galeani et al., 2009]. Nearest in the other direction, Gao et al. [2025] run the same recursion but gate the *model* update on the conformal score. Their one non-standard assumption is that $\sum_{t \le T} q_t \le O(\alpha\, h(T))$ for a sublinear h, supported with a plot rather than derived.

5 Limitations

The evidence is one construction, one adversary, one saturator; and $0.63/(1-w)$ is specific to the saturator's *level*, which condition (4) bounds only from below. **The inherited guarantee is weak where it applies.** Under the tangent integrator $h(t) = t/\log t$, the certified gap $(c\,h(T)+1)/T$ decays as $O(1/\log T)$ rather than $O(1/T)$. What is forfeited was not a tight certificate. **The dead-band widths are absolute, and the failure is total only against the adversary.** τ is in score units with $b = 2$, so the failing $\tau = 1.5$ is $0.75\,b$. Under i.i.d. scores it returns 0.249376 at $T = 10^6$, an adversary-free confirmation of the mechanism rather than a retreat. The threshold pins at $b - \tau = 0.5$, and uniform scores on $[-b/2, b/2]$ put $P(s > 0.5) = 0.25$ exactly. **The harness was rebuilt against numbers already read**, which carries less weight than a rerun done without sight of them. **Placement B is conceded, not tested. One seed, and only the null scorecaster's edge is bracketed**; the others are one-sided. **What would overturn the claim** is a legal $(r_t, \hat{q})$ pair with $\tau \le \tau^\star$ that loses coverage, or $\tau > \tau^\star$ that keeps it; the sweep has neither.

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A Supporting measurements

The primary regime. Every figure in Table 1 is measured under one deterministic adversary. It plays $s_t = \mathrm{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$ with $\varepsilon = 10^{-9}$ against the *deployed* threshold, and the indicator is strict. The specification is not a tie rule, and the distinction is measured rather than argued. Scoring an exact-threshold play *as a miscoverage* reproduces the primary regime on every cell of Table 1: 44 of 44 identical, all eleven arms at all four horizons. Scoring the same play under the *strict* indicator does not, and it moves every row, because the adversary's signature move then covers rather than misses: partial adjustment at $w = 0.999$ excurses 70.80 in place of 623.70, and the failing dead band returns 0.000000 in place of 1.000000. The $\varepsilon \downarrow 0$ limit the harness specifies is the first reading. **No row is convention-free with respect to the second**, which is why the adversary is specified rather than tie-broken.

Excursions at $T = 10^4$, and the law across the arms. Partial adjustment excurses 6.30, 63.00 and 623.70 at $w = 0.9$, 0.99 and 0.999. The last two are flat in T while Proposition 2's bound grows like $\log T$, which is why the ratio falls with the horizon. Across the arms the excursion fits $\max_t |E_t| \approx \max\{0.5\, h(T) + 0.9,\ 0.63/(1-w)\}$. This is a fit to the measured arms, not a derived bound.

Realised miscoverage at the covering settings of Figure 2. At $\hat{q} \equiv +b/2$ the $\tau = 1.5$ band returns 0.100010 with $\max_t |E_t| = 11.20$, inside the bound. Under Angelopoulos et al.'s tangent integrator the same width returns 0.100001. Raising the saturator's level to $4b$ leaves coverage intact and cuts the $w = 0.999$ excursion from 623.70 to 120.60, which is the first term of $\tau^\star$ moving the edge rather than the readout changing behaviour.

B The four vocabularies

Table 2: Four literatures name the movement of a sequentially updated forecast in four vocabularies: *abrupt threshold changes*; *forecast (in)stability* (MASC, MAC/SDC, MQC/SMQC, congruence, nervousness); *jumpiness*, *convergence index*, *(in)consistency*; and *switching cost*. The row gives what each contributes.

Online conformal	**Forecast stability**	**Verification**	**Switching-cost online learning**
a distribution-free long-run-coverage guarantee for *any* bounded scorecaster, with a rate [Angelopoulos et al., 2023, Thm 1, Prop. 2], and the admissible set measured here to be tight	the scored movement functional, the readout map as post-processing [Godahewa et al., 2025, Eq. 3], and a priced decision [Genov et al., 2026, §4.2] and [Van Belle et al., 2026]	the matched-level design on real producers from 2012, with a control *stronger* than the pair: the whole marginal [Pinson and Girard, 2012, Worsnop et al., 2018]	worst-case theory for the penalised problem: Thm 2, and a Thm 7 trade-off in which sublinear regret is bought by growing θ with T, which grows the ratio [Andrew et al., 2013]